

DHARSAN KESAVAN

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EDUCATION

Arizona State University, Tempe, AZ

Expected May 2030

Dual B.S., Biological Sciences (Biomedical Sciences) and AI in Business

- McKenna Life Sciences, Business and Entrepreneurship (MLSBE), **inaugural cohort**. A 30-student program run jointly by W. P. Carey and The College: two bachelor's degrees plus an Entrepreneurship and Innovation certificate in four years.

BASIS Chandler, Chandler, AZ

Aug 2022 – May 2026

- AP Scholar with Distinction. Scores of 5 in AP Biology, AP Chemistry, and AP Physics 1. SAT 1530.

VENTURES AND PRODUCTS

Cognifer Labs, Co-Founder

Mar 2026 – Present

Avorio, a native spaced-repetition learning app for macOS and iOS. avorio.ai

- Shipped Avorio to the App Store — version 1.0 approved July 2026, and 1.1 approved and released the same day it was submitted — with the Mac build available as a signed, notarized direct download that updates itself.
- Designed one core so every platform behaves identically instead of three implementations quietly drifting apart: a single engine handles scheduling, storage, Anki import/export, gamification, and AI routing, and both native UIs call straight into it.
- Made switching from Anki lossless: importing a deck keeps its full history — ease factors, lapses, media, cloze, image occlusion — behind a review step before anything commits, then re-fits the learner's own forgetting-curve parameters from that history instead of starting them cold.
- Shipped the AI layer with a free choice of model and free on-device paths, so cost is never the reason a learner can't use it: turns a document into cards and diagrams, explains a single card, and tutors toward whatever point the learner keeps missing.
- Ran the backend and the business model: a free flashcard core, paid tiers for cloud AI and cross-device sync, checkout and subscriptions live on web and both app stores. Localized into eight languages, with an automated test suite gating every change, and Android in progress on the same core.

BioPath, Founder and Sole Engineer

2025 – Present

Drug interaction checker and compound body-impact analysis. biopath.space

- Built a computational pharmacology tool that answers one question end to end: what will this compound do inside a body, and is it safe with what someone already takes. It starts from a drug name, plant photo, plant name, or pill imprint.
- Connected ten scientific databases, each holding one piece of the answer and none of the others (PubChem, ChEMBL, Reactome, Open Targets, DGIdb, OpenFDA, RxNorm, PlantNet, Dr. Duke's). The pipeline resolves identity, finds binding targets, maps them onto 200+ pathways, scores organ impact, and returns interaction, dosage, and pregnancy safety.
- Trained a Morgan-fingerprint CNN and MPNN on BindingDB to predict binding affinity across roughly 700 human proteins, covering obscure natural products and novel molecules with no measured data. Predictions are cross-checked against measured ChEMBL targets: agreement raises confidence, disagreement is flagged rather than hidden.
- Used Tanimoto structural similarity at a 0.70 threshold to carry side effects, pregnancy categories, and interaction risk over from analogous known drugs. Every prediction shows its provenance and confidence rather than a bare number.

AlóJefe, Full-Stack Developer (client)

2025 – 2026

- Built the site and backend for a bilingual AI phone assistant sold to solo contractors in roofing, HVAC, and plumbing. It answers the calls an owner cannot take from a roof, qualifies the job, books an estimate with drive-time routing, and texts back a summary, switching between English and Spanish mid-sentence. Next.js on Vercel.

CLINICAL EXPERIENCE

Banner Health, Patient-Facing Volunteer

2024 – Present · 175 hours

- Assisted patients and clinical staff across inpatient and outpatient departments at one of the largest nonprofit health systems in the country. Built a front-desk data system that cut the time it took to log and retrieve patient information.

Desert Kidney Associates, Physician Shadowing, Nephrology

2024 – Present · 120 hours

- Shadowed Dr. Prashant Kolar through inpatient rounds, outpatient consultations, and dialysis management. Followed chronic kidney disease staging, fluid and electrolyte management, and the way diabetes and hypertension end up as renal outcomes. Sat in on patient education about lifestyle change, which is where much of the disease course actually gets decided.

ROC Physical Therapy, Clinical Shadowing

2023 · 80 hours

- Observed outpatient rehabilitation with physical therapist Salman Ashraf: gait and movement assessment, therapeutic exercise progression, manual therapy, and the arc from injury back to function.

Emergency Medical Technician, training in progress

2026

RESEARCH AND SCHOLARLY WRITING

- "Non-Pharmacological Therapies for Alzheimer's Disease: MST and CST," *literature review* Sept 2025
- Wrote a literature review for *Mindful Mental Health* on Medium comparing Mnemonic Strategy Training and Cognitive Stimulation Therapy. Worked through the neuroimaging evidence and the published effect sizes for each, showed why the two cannot be compared head-to-head — different outcome measures, different patient populations — and argued they are complementary rather than competing.

LEADERSHIP AND TEACHING

- Flare Review**, *Director of Operations* Oct 2025 – Present
- Run operations for a student-led video editing nonprofit: scheduling, project intake, and hand-off between editors.
- Schoolhouse.world**, *Senior Tutor* Nov 2023 – Jun 2026
- Tutored one on one, building each plan around what the student was actually getting wrong. Recruited and onboarded new tutors, which is what let the program reach more students than any one person could.
- Volunteer Teacher**, *student meal program* 2023 – 2025
- Raised \$7,000 for student meal programs and taught in the classrooms that money paid for. Also coached students on career paths, the part they asked for most.
- Arizona Tamil School**, *Teaching Assistant* 10+ years
- Taught language and culture classes for a decade and coordinated the annual performances, from rehearsal scheduling through the night itself.

OTHER SOFTWARE

- Agent Bridge.** A Tauri desktop app in Rust and React that drives Claude Code, Codex, and Cursor from one shell over the Agent Client Protocol. One canonical config projects into each tool's native format and round-trips back unchanged, and switching agents rebuilds context under a label reading "reconstructed, not resumed" rather than pretending the session moved. 10,400 lines, 117 tests.
- Backglass.** A local-first commitment ledger in Python (102 modules, 20 source connectors, 71 test files) that reads Gmail, Drive, Calendar, notes, iMessage, Slack, and GitHub on a schedule and extracts typed records: what you owe, what is owed to you, deadlines, and goal checkpoints, each linked back to the source item it came from. It returns a 6 a.m. brief and a dashboard, with no chat interface and no vector database on purpose: the questions are fixed and known in advance, so the work belongs at ingest in plain SQL. A cheap triage tier sits in front of the model under a hard monthly spend cap, and across 3,687 real messages a zero-cost pre-filter removed 10.9% of the volume with zero disagreement against the model verdicts it replaced.
- Misinformation contagion simulators.** Two agent-based models of how a claim spreads through a synthetic Phoenix. The C++ and SFML build runs 10,000 agents over 690 steps on an SEDPNR state model, with contact probability weighted by shared school, workplace, age, and ethnicity, and a false claim carrying the six-fold transmission advantage Vosoughi and colleagues measured in *Science* in 2018. A second model in Python gives each agent dual-process System 1 and System 2 reasoning, identity-threat detection, and a demographically biased media diet, so adoption depends on who the agent is and not only on how many neighbours already believe it. Both write per-timestep state counts for analysis, and the C++ build ships a live SFML visualizer.

HONORS AND CERTIFICATIONS

Honors	Best Delegate, Model United Nations. One gold and three silver medals, Academic Decathlon state competition.
Certifications	Introductory Human Physiology, Duke University (Aug 2025, credential 3SYLO9PVP12V). Machine Learning in Healthcare, Northeastern University. OSHA Bloodborne Pathogens and HIPAA for healthcare workers.

SKILLS

Biomedical	Pharmacology: drug mechanisms, receptor binding, ADME, drug-drug interactions. Bioinformatics databases: PubChem, ChEMBL, Reactome, Open Targets, DGIdb, OpenFDA, RxNorm, DrugBank. Renal physiology. Literature review and effect-size interpretation.
Programming	Rust, Swift and SwiftUI, Kotlin and Jetpack Compose, Python, TypeScript, C++. UniFFI, rusqlite, React and Next.js, Tauri.
Machine learning	Morgan-fingerprint CNN and MPNN training on BindingDB, cross-validation against measured data, Tanimoto similarity, LLM integration through Anthropic, OpenAI, and Ollama.
Infrastructure	Supabase and Postgres, SQLite in WAL mode, Deno edge functions, Vercel, Stripe, RevenueCat and StoreKit, GitHub Actions CI.
Business	Product pricing and tiering, App Store submission and review, code signing and notarization, localization, freelance delivery, fundraising, recruiting.